

IPv4 and Binary — My Learning Notes So Far

1. What is a Bit?

A **bit** is the smallest unit of information in binary.

A bit can contain only:

0
or
1

Think of a bit as one box:

[0]

or:

[1]

Therefore:

1 box = 1 bit

Examples:

[1]	= 1 bit
[1][0]	= 2 bits
[1][1][0]	= 3 bits
[1][0][1][1]	= 4 bits

My understanding

I was initially confused about what a bit actually was.

I learned:

A bit is one position that can contain either **0** or **1**.

2. What Are Multiple Bits?

If we have two bits:

```
[ ] [ ]
```

each box can contain either `0` or `1`.

The possible 2-bit values are:

```
00
01
10
11
```

There are 4 different values.

For 3 bits:

```
000
001
010
011
100
101
110
111
```

There are 8 different values.

The important idea is:

We are not adding another bit when we add another line. The number of bits stays the same; we are simply listing the different ways those bits can contain `0` and `1`.

For example:

```
000 → 3 bits
001 → 3 bits
010 → 3 bits
111 → 3 bits
```

They are all 3 bits.

3. Why Does the Number of Values Double?

Every bit can be either:

0
or
1

Therefore:

1 bit → 2 values
2 bits → 4 values
3 bits → 8 values
4 bits → 16 values
5 bits → 32 values
6 bits → 64 values
7 bits → 128 values
8 bits → 256 values

The important pattern is:

Every time we add 1 bit, the number of possible values doubles.

For example:

2 → 4 → 8 → 16 → 32 → 64 → 128 → 256

4. Powers

A power is simply a shortcut for repeated multiplication.

For example:

2^3

means:

$2 \times 2 \times 2$

Therefore:

$$2^3 = 8$$

Similarly:

$$2^5$$

means:

$$2 \times 2 \times 2 \times 2 \times 2$$

and:

$$2^5 = 32$$

The small number tells us how many times we multiply the number by itself.

So:

$$\begin{aligned} 2^1 &= 2 \\ 2^2 &= 4 \\ 2^3 &= 8 \\ 2^4 &= 16 \\ 2^5 &= 32 \\ 2^6 &= 64 \\ 2^7 &= 128 \\ 2^8 &= 256 \end{aligned}$$

For networking:

$$2^8 = 256$$

means 8 bits can represent **256 different values**.

Those values are:

0 through 255

5. 8-Bit Binary

An 8-bit binary number has 8 positions.

The position values are:

128 64 32 16 8 4 2 1

For example:

10100000

means:

128	64	32	16	8	4	2	1
1	0	1	0	0	0	0	0

So:

$128 + 32 = 160$

Therefore:

$10100000 = 160$

I practiced several binary-to-decimal conversions and became comfortable with them.

6. Why an 8-Bit Number Goes From 0 to 255

There are:

$2^8 = 256$

different values possible with 8 bits.

Because we start counting from 0, the values are:

0 → 255

Therefore:

8 bits = 256 possible values
0 through 255

The largest 8-bit binary number is:

11111111

Its decimal value is:

$128 + 64 + 32 + 16 + 8 + 4 + 2 + 1$
 $= 255$

7. What Is an IPv4 Address?

An IPv4 address contains **32 bits**.

IPv4 is written as four sections called **octets**.

Example:

192.168.1.1

Each octet contains 8 bits:

192	.	168	.	1	.	1
octet		octet		octet		octet
8 bits		8 bits		8 bits		8 bits

Therefore:

$8 + 8 + 8 + 8 = 32 \text{ bits}$

So:

IPv4 = 32 bits = 4 octets = 8 bits per octet.

8. Example: 192.168.1.1 in Binary

We converted:

```
192 → 11000000
168 → 10101000
1   → 00000001
1   → 00000001
```

Therefore:

```
192.168.1.1

11000000.10101000.00000001.00000001
```

Each section is exactly 8 bits.

9. What Does `/24` Mean?

Consider:

```
192.168.1.100/24
```

The `/24` means:

```
24 network bits
```

Since IPv4 has 32 total bits:

```
32 - 24 = 8
```

Therefore:

```
24 network bits
8 host bits
```

Because each octet is 8 bits:

```
8 + 8 + 8 = 24
```

Therefore `/24` means:

First 3 octets = network
Last octet = host

Example:

192.168.1.100/24

Network = 192.168.1
Host = 100

10. `/24` Address Calculation

For `/24`:

32 total bits
-24 network bits

8 host bits

We know:

$$2^8 = 256$$

Therefore:

256 total addresses

Normally two addresses are reserved:

Network address
Broadcast address

So:

$$256 - 2 = 254$$

Therefore:

/24 normally has **254 usable host addresses**.

11. Network Address

The **network address** identifies the network itself.

For:

172.16.45.123/24

The network portion is:

172.16.45

The host portion is:

123

To find the network address, set the host portion to **0**:

172.16.45.0

Therefore:

Network address = 172.16.45.0

12. Broadcast Address

The broadcast address is the **last address in the network**.

For /24, the host portion has 8 bits.

The maximum 8-bit value is:

255

Therefore:

172.16.45.255

is the broadcast address.

So:

Network address:
172.16.45.0

Broadcast:
172.16.45.255

Usable:
172.16.45.1 → 172.16.45.254

13. My /24 Exercise

Question:

172.16.45.123/24

My answer:

Network part: 172.16.45
Host part: 123

Initially, I answered:

244

for usable devices.

That was incorrect.

Correct calculation:

8 host bits
→ 2^8
→ 256 total addresses

256 - 2
= 254 usable addresses

Correct answer:

254 usable devices

14. Another /24 Exercise

Question:

10.25.7.200/24

My answer:

Network part: 10.25.7
Host part: 200
Network bits: 24
Host bits: 8
Usable devices: 254

This was correct.

Network address:

10.25.7.0

Broadcast:

10.25.7.255

Usable:

10.25.7.1 → 10.25.7.254

15. What Does /16 Mean?

Consider:

10.25.100.50/16

The /16 means:

16 network bits

IPv4 has 32 bits:

$32 - 16 = 16$

Therefore:

16 network bits
16 host bits

Since each octet has 8 bits:

$8 + 8 = 16$

Therefore:

First 2 octets = network
Last 2 octets = host

For:

10.25.100.50/16

we have:

Network = 10.25
Host = 100.50

16. /16 Address Calculation

There are 16 host bits.

Therefore:

$$2^{16}$$

means multiplying 2 by itself 16 times:

$$\begin{array}{l} 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \\ \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \end{array}$$

The result is:

65,536

Therefore:

16 host bits
→ 65,536 total addresses

Remove:

1 network address
1 broadcast address

Therefore:

$$\begin{array}{l} 65,536 - 2 \\ = 65,534 \text{ usable host addresses} \end{array}$$

17. /16 Network Address

For:

10.20.45.100/16

The network portion is:

10.20

The host portion is:

45.100

To find the network address, make the entire host portion **0**:

10.20.0.0

Important:

Because **/16** has TWO host octets, BOTH must become zero.

Incorrect:

10.20.45.0

Correct:

10.20.0.0

18. **/16** Broadcast Address

The host portion is:

45.100

There are two host octets.

The largest value of each 8-bit octet is **255**.

Therefore:

10.20.255.255

is the broadcast address.

So:

Network:
10.20.0.0

Usable:

10.20.0.1 → 10.20.255.254

Broadcast:
10.20.255.255

19. Why Does /16 Have a Large Usable Range?

This confused me at first.

For:

10.20.45.100/16

we have:

10.20		45.100
NETWORK		HOST

The two host octets can vary.

The host portion can start at:

0.0

and go all the way to:

255.255

Therefore the full range is:

```
10.20.0.0
10.20.0.1
10.20.0.2
...
10.20.0.255
10.20.1.0
10.20.1.1
...
10.20.255.254
10.20.255.255
```

The first address is reserved for the network:

10.20.0.0

The last address is reserved for broadcast:

10.20.255.255

Therefore usable addresses are:

10.20.0.1 → 10.20.255.254

Important:

65,534 is the **number of usable addresses**.

It is NOT an IP address.

An IP octet can only be:

0 → 255

20. My /16 Exercise

Question:

172.20.45.123/16

My first answer was:

Network: 172.20.45
Host: 123

This was wrong because I was thinking about /24.

Correct:

Network: 172.20
Host: 45.123

Then:

Network address:

172.20.0.0

Broadcast:

172.20.255.255

Usable:

172.20.0.1 → 172.20.255.254

Usable devices:

65,534

21. My Correct /16 Exercise

Question:

10.50.123.45/16

My answer:

Network part: 10.50

Host part: 123.45

Network bits: 16

Host bits: 16

Total addresses: 65,536

Usable devices: 65,534

This was correct.

Network address:

10.50.0.0

Broadcast:

10.50.255.255

Usable:

10.50.0.1 → 10.50.255.254

22. Mixed /16 and /24 Practice

I successfully solved:

```
192.168.50.75/24
```

Answer:

```
Network: 192.168.50
Host: 75

Network address:
192.168.50.0

Broadcast:
192.168.50.255

Usable:
192.168.50.1 → 192.168.50.254

Usable devices:
254
```

I also solved:

```
10.40.75.120/16
```

Correct:

```
Network: 10.40
Host: 75.120

Network:
10.40.0.0

Broadcast:
10.40.255.255

Usable:
10.40.0.1 → 10.40.255.254

Usable devices:
65,534
```

I made one mistake initially by writing:

10.40.65534.65534

This was wrong because 65,534 is a **count**, not an IP octet.

Each IP octet can only contain:

0 → 255

23. Another Correct /24

Question:

172.25.100.200/24

My answer:

Network: 172.25.100

Host: 200

Network:

172.25.100.0

Broadcast:

172.25.100.255

Usable:

172.25.100.1 → 172.25.100.254

Usable devices:

254

Correct.

24. Another Correct /16

Question:

10.15.200.50/16

My answer:

Network: 10.15

Host: 200.50

Network:

10.15.0.0

Broadcast:

10.15.255.255

Usable:

10.15.0.1 → 10.15.255.254

Usable devices:

65,534

Correct.

25. The Most Important Rules I Learned

Rule 1 — IPv4

IPv4 = 32 bits

It has:

4 octets

Each octet:

8 bits

Rule 2 — Each bit

1 bit = 0 or 1

Rule 3 — Number of values

1 bit → 2 values
2 bits → 4 values
3 bits → 8 values
4 bits → 16 values
5 bits → 32 values
6 bits → 64 values
7 bits → 128 values
8 bits → 256 values

Every additional bit **doubles** the number of values.

Rule 4 — Powers

$$2^3 = 2 \times 2 \times 2 = 8$$

The small number tells how many times to multiply.

For example:

$$2^8 = 256$$

and:

$$2^{16} = 65,536$$

Rule 5 — /24

/24

24 network bits
8 host bits

Therefore:

3 network octets
1 host octet

Total:

$2^8 = 256$ addresses

Usable:

$256 - 2 = 254$

Rule 6 — /16

/16

16 network bits

16 host bits

Therefore:

2 network octets

2 host octets

Total:

$2^{16} = 65,536$ addresses

Usable:

$65,536 - 2 = 65,534$

26. Network Address Rule

Set all **host bits** to 0.

Example /24:

192.168.1.100/24

Host:

100

Network:

192.168.1.0

Example **/16**:

10.20.45.100/16

Host:

45.100

Network:

10.20.0.0

27. Broadcast Address Rule

Set all **host bits to 1**.

For an 8-bit host portion:

11111111 = 255

For **/24**:

192.168.1.255

For **/16**:

10.20.255.255

28. Usable Host Rule

Normally:

Network address 
Broadcast address 
Everything between them 

For `/24`:

Network: .0
Usable: .1 → .254
Broadcast: .255

For `/16`:

Network: .0.0
Usable: .0.1 → .255.254
Broadcast: .255.255

29. What Is Subnetting?

After learning `/16` and `/24`, we started subnetting.

The basic idea:

Subnetting means taking one big network and dividing it into smaller networks.

Example:

One big network:

192.168.1.0/24

Can be divided into smaller networks.

For example:

192.168.1.0/25
192.168.1.128/25

So:

```
ONE BIG NETWORK
  ↓
SUBNETTING
  ↓
SMALLER NETWORKS
```

30. Why Does `/24` Become `/25`?

This was one of my important questions.

I asked:

"Where did we take 1 and add it to 24 to get 25?"

The answer:

We did NOT add a new bit.

We **borrowed 1 bit from the host portion.**

Before:

```
/24

24 network bits
8 host bits
```

After borrowing one host bit:

```
/25

25 network bits
7 host bits
```

So:

```
/24 → /25
```

means:

24 network bits + 8 host bits

becomes:

25 network bits + 7 host bits

The total is still:

32 bits

We didn't create an extra bit.

We changed one host bit into a network bit.

31. Why Does Borrowing 1 Bit Create 2 Subnets?

One bit can contain:

0
or
1

Therefore the borrowed bit gives us two groups:

borrowed bit = 0
→ subnet 1

borrowed bit = 1
→ subnet 2

Therefore:

Borrow 1 bit
→ 2 subnets

This is why:

/24 → /25

creates two subnets.

32. /24 to /25 Example

Original:

```
192.168.1.0/24
```

We have:

```
24 network bits  
8 host bits
```

Binary structure:

```
11111111.11111111.11111111.00000000  
<----- network -----><--host-->
```

Borrow one host bit:

```
11111111.11111111.11111111.10000000  
<----- 25 network -----><7 host>
```

Now:

```
25 network bits  
7 host bits
```

Since:

```
 $2^7 = 128$ 
```

each subnet has:

```
128 total addresses  
126 usable addresses
```

33. The Two /25 Subnets

The original /24 :

192.168.1.0 → 192.168.1.255

is divided into:

Subnet 1

192.168.1.0/25

Network:

192.168.1.0

Usable:

192.168.1.1 → 192.168.1.126

Broadcast:

192.168.1.127

Subnet 2

192.168.1.128/25

Network:

192.168.1.128

Usable:

192.168.1.129 → 192.168.1.254

Broadcast:

192.168.1.255

So:

Original:

192.168.1.0/24

↓

split

↓

192.168.1.0/25

192.168.1.128/25

34. My Question About the Third Octet

I asked:

"But you split; still the 3rd octet remains 1?"

The answer:

Yes.

For:

192.168.1.0/24

the third octet **1** is already part of the network because:

8 + 8 + 8 = 24

So **/24** means:

192.168.1		00000000
^^^^^^^^^^		^^^^^^^^^^
NETWORK		HOST

When we change **/24** to **/25**, we borrow from the **fourth octet**, not the third.

Therefore the third octet remains:

1

The fourth octet changes.

35. The 128, 64, 32, 16, 8, 4, 2, 1 Positions

Every 8-bit octet always has these positions:

128 64 32 16 8 4 2 1

These values do NOT move or disappear.

Before **/25**:

128	64	32	16	8	4	2	1
H	H	H	H	H	H	H	H

All 8 positions are host bits.

After /25 :

128	64	32	16	8	4	2	1
N	H	H	H	H	H	H	H

The 128 position still exists.

The only thing that changed is its **role**:

Before:

128 = host bit

After:

128 = network bit

The other positions remain host bits:

64	32	16	8	4	2	1
----	----	----	---	---	---	---

36. Why Do /25 Subnets Start at .0 and .128?

The borrowed bit is the 128 position.

That bit can be:

0

or:

1

If it is 0 :

00000000

The fourth octet starts at:

0

So:

192.168.1.0/25

If it is **1**:

10000000

The **1** is in the **128** position:

128	64	32	16	8	4	2	1
1	0	0	0	0	0	0	0

Therefore:

10000000 = 128

So the second subnet starts at:

192.168.1.128/25

37. Current Learning Position

So far, I have learned:

```
Binary
  ↓
Bits
  ↓
8-bit values
  ↓
IPv4 = 32 bits
  ↓
```

```
4 octets
↓
/24
↓
/16
↓
Network and host portions
↓
Network address
↓
Broadcast address
↓
Usable host range
↓
Subnetting
↓
/24 → /25
```

The next thing I need to fully understand is:

How subnetting changes the 128/64/32/16/8/4/2/1 positions

especially how the borrowed bit creates subnet boundaries.

38. My Quick Reference

```
IPv4
= 32 bits
= 4 octets
= 8 bits per octet
```

```
8 bits
= 256 values
= 0 → 255
```

```
/24
= 24 network bits
= 8 host bits
= 256 total addresses
= 254 usable
```

```
/16
= 16 network bits
```

= 16 host bits
= 65,536 total addresses
= 65,534 usable

Network address
= host bits all 0

Broadcast address
= host bits all 1

Usable hosts
= between network and broadcast

Subnetting
= borrowing host bits
and turning them into network bits

Borrow 1 bit
→ 2 subnets

/24 → /25
24 network bits → 25 network bits
8 host bits → 7 host bits

$$2^7 = 128$$

So each /25 has:

128 total addresses
126 usable

And:

192.168.1.0/25
192.168.1.128/25

are the two /25 subnets created from:

192.168.1.0/24

Final Mental Model

The most important thing I understand now is:

An IP address is 32 bits. The `/number` tells me how many of those 32 bits belong to the network. The remaining bits belong to hosts.

For example:

```
/24
24 network + 8 host

/16
16 network + 16 host

/25
25 network + 7 host
```

And subnetting works by:

Taking some bits from the host side and giving them to the network side.